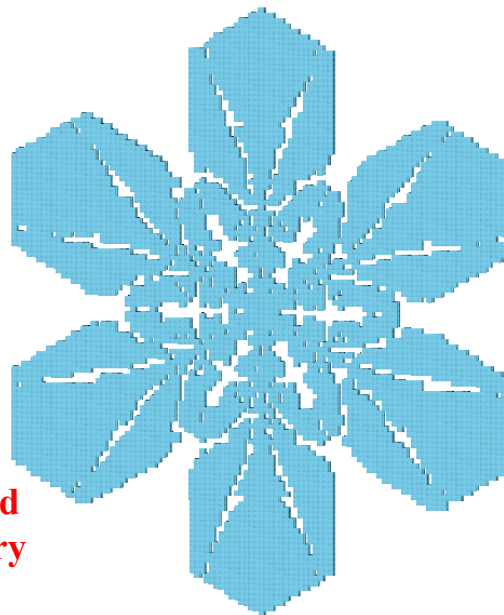


MIDAS: MoM Integral-equation Decomposition for Arbitrarily-shaped Scatterers

3D full-wave model, based on the volume integral equation method (VIEM)

$$\bar{E}(\bar{r}) = \bar{E}^i(\bar{r}) + (k_0^2 + \nabla \nabla \cdot) \int_{\Omega} \chi(\bar{r}') G(\bar{r}, \bar{r}') E(\bar{r}') d\bar{r}'$$

where $\bar{E}(\bar{r})$ is the field inside the scatterer, $\bar{E}^i(\bar{r})$ is the incident field, $\chi(\bar{r}')$ is the dielectric contrast at the location r' , k_0 is the wavenumber in air and $G(\bar{r}, \bar{r}')$ is the free space Green's function.



Pristine crystal
particle discretized
into Nb_c elementary
cubic cells

$$\bar{\Gamma} \bar{E}(\bar{r}) = \bar{E}^{ref}(\bar{r})$$

Method of Moments

The particle is discretized into Nb_c cubic cells Ω_n , of side S_c

$$S_c \leq \frac{\lambda_s}{10}; \lambda_s = \frac{\lambda_0}{\sqrt{Re(\epsilon_r)}}$$

Use of piecewise
constant basis functions

$$\bar{E}(\bar{r}) = \sum_{n=1}^N \sum_{q=1}^3 E_q^n \bar{F}_q^n(\bar{r})$$

Direct
solver

$$\mathbf{Z} \mathbf{E} = \mathbf{E}^{inc}$$



MIDAS: MoM Integral-equation Decomposition for Arbitrarily-shaped Scatterers

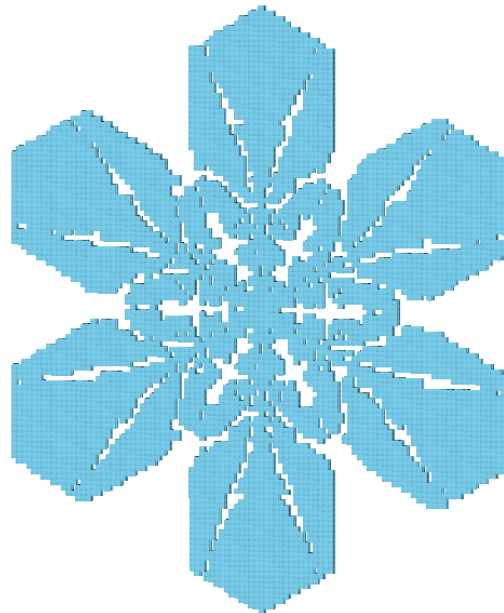
3D full-wave model, based on the volume integral equation method (**VIEM**)

$$\bar{E}(\bar{r}) = \bar{E}^i(\bar{r}) + (k_0^2 + \nabla \nabla \cdot) \int_{\Omega} \chi(\bar{r}') G(\bar{r}, \bar{r}') E(\bar{r}') d\bar{r}'$$

Storage of the entire matrix is prohibitive for direct methods



Direct solver



$$\bar{\Gamma} \bar{E}(\bar{r}) = \bar{E}^{ref}(\bar{r})$$

Method of Moments

The particle is discretized into **Nbc** cubic cells Ω_n , of side S_c

$$S_c \leq \frac{\lambda_s}{10}; \lambda_s = \frac{\lambda_0}{\sqrt{Re(\epsilon_r)}}$$

Use of piecewise constant basis functions

$$\bar{E}(\bar{r}) = \sum_{n=1}^N \sum_{q=1}^3 E_q^n \bar{F}_q^n(\bar{r})$$

$$\mathbf{Z} \mathbf{E} = \mathbf{E}^{inc}$$



MIDAS: MoM Integral-equation Decomposition for Arbitrarily-shaped Scatterers

3D full-wave model, based on the volume integral equation method (**VIEM**)

$$\bar{E}(\bar{r}) = \bar{E}^i(\bar{r}) + (k_0^2 + \nabla \nabla \cdot) \int_{\Omega} \chi(\bar{r}') G(\bar{r}, \bar{r}') E(\bar{r}') d\bar{r}'$$

$$\bar{\Gamma} \bar{E}(\bar{r}) = \bar{E}^{ref}(\bar{r})$$

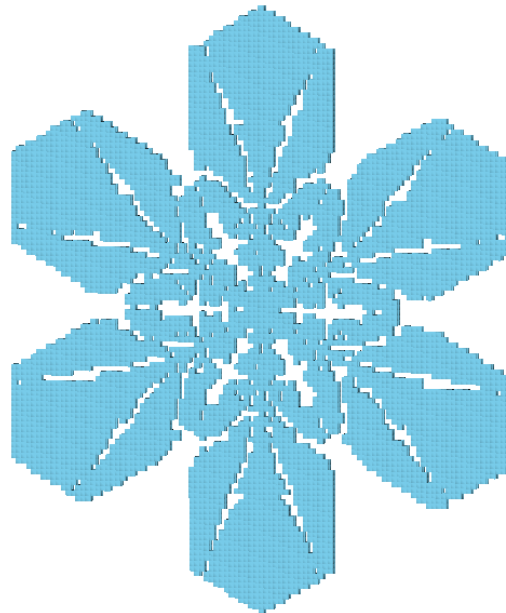
Method of Moments

The particle is discretized into N cubic cells Ω_n , of side d

$$d \leq \frac{\lambda_s}{10}; \lambda_s = \frac{\lambda}{\sqrt{\mathcal{R}(\epsilon_r)}}$$

Use of piecewise constant basis functions

$$\bar{E}(\bar{r}) = \sum_{n=1}^N \sum_{q=1}^3 E_q^n \bar{F}_q^n(\bar{r})$$



Direct
solver



CBFM

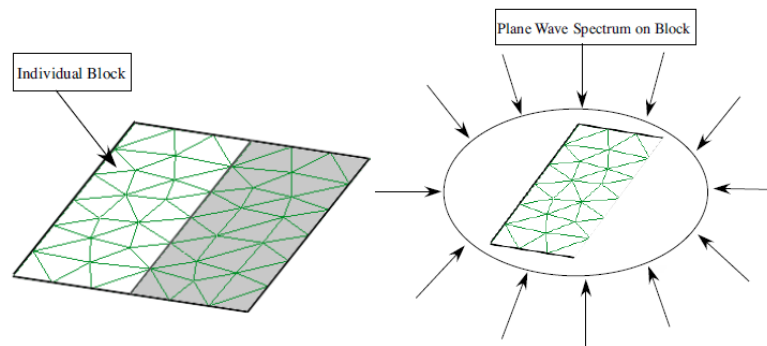


$$\mathbf{Z} \mathbf{E} = \mathbf{E}^{inc}$$

MIDAS: MoM Integral-equation Decomposition for Arbitrarily-shaped Scatterers

Direct solver-based

- ❖ Better adapted to **multiple right-hand** side problem
- ❖ Subject to a wide variety of enhancement techniques
- ❖ Tunable depending on to the needs (memory or CPU) through the size of the blocks (h_B or $N_{b,max}$).
- ❖ Highly amenable to **MPI** parallelization

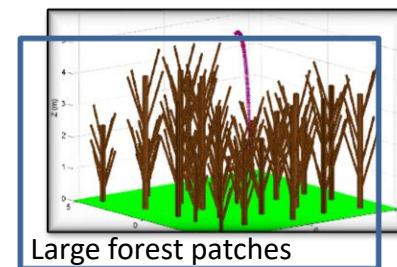
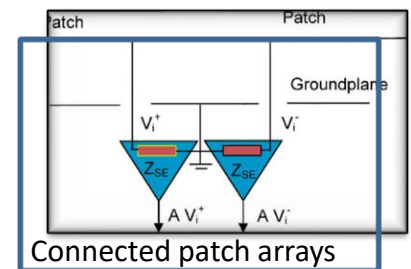
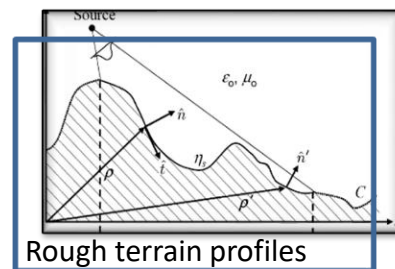
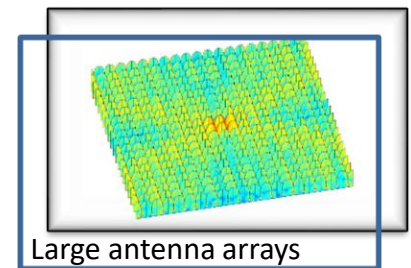
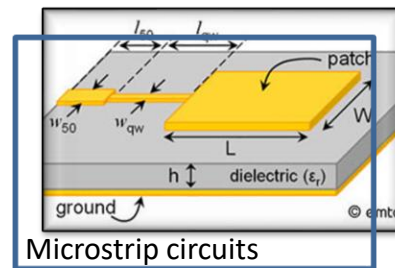


Spectrum of plane waves incident on a single block

R. Maaskant, R. Mittra, A. Tijhuis, "Fast Analysis of Large Antenna Arrays Using the Characteristic Basis Function Method and the Adaptive Cross Approximation Algorithm", IEEE Transactions on Antennas and Propagation, 2008.

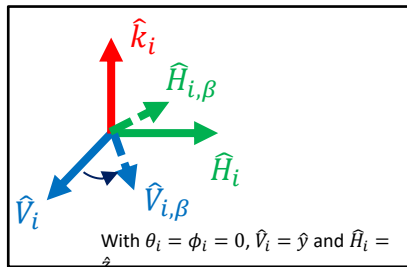
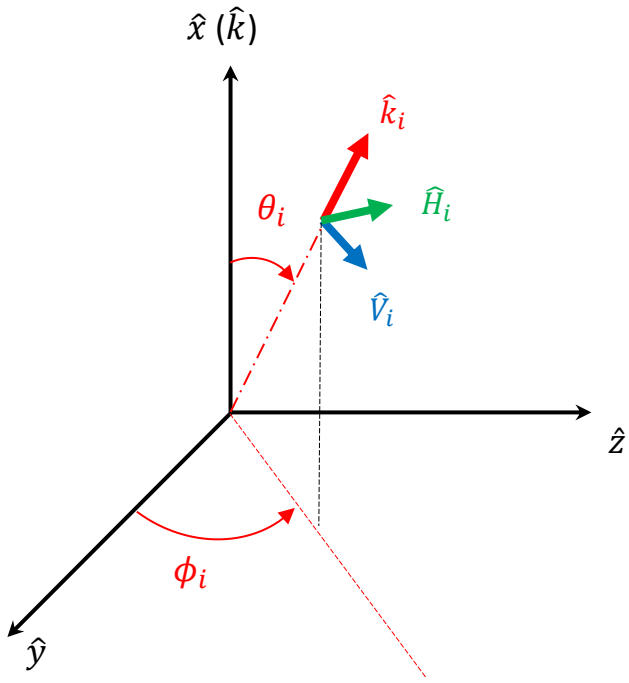
Jaime Laviada et al, "Solution of Electrically Large Problems With Multilevel Characteristic Basis Functions", IEEE Trans. on Antennas Propagation, 2009.

The **CBFM** is a powerful **Domain Decomposition method (DDM)** which has been proven to be accurate and efficient when applied to large-scale EM problems, even when the computational resources are limited



MIDAS Incident direction and polarization rotation VS DDSCAT target orientation definitions

MIDAS



The orientation of the incident wave vector $\hat{k}_i$, and thus of the orthonormal basis vectors $\hat{v}_i$ and $\hat{h}_i$, is specified by angles $\theta_i \in [0 \pi]$ and $\phi_i \in [0 2\pi]$. This translates into:

$$\hat{k}_i = \cos \theta_i \hat{x} + \sin \theta_i \cos \phi_i \hat{y} + \sin \theta_i \sin \phi_i \hat{z}$$

$$\hat{v}_i = -\sin \theta_i \hat{x} + \cos \theta_i \cos \phi_i \hat{y} + \cos \theta_i \sin \phi_i \hat{z}$$

$$\hat{h}_i = -\sin \phi_i \hat{y} + \cos \phi_i \hat{z}$$

Following the Forward Scattering Alignment (FSA) convention, the same equations apply to the scattered wave and polarizations directions

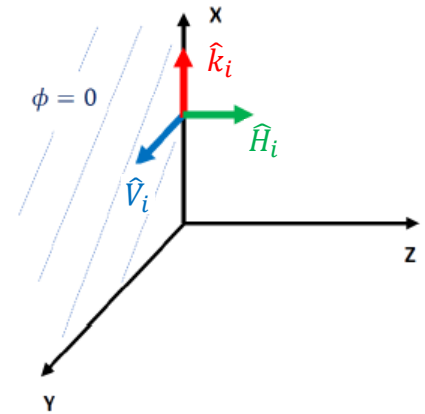
$$\hat{k}_s = \cos \theta_s \hat{x} + \sin \theta_s \cos \phi_s \hat{y} + \sin \theta_s \sin \phi_s \hat{z}$$

$$\hat{v}_s = -\sin \theta_s \hat{x} + \cos \theta_s \cos \phi_s \hat{y} + \cos \theta_s \sin \phi_s \hat{z}$$

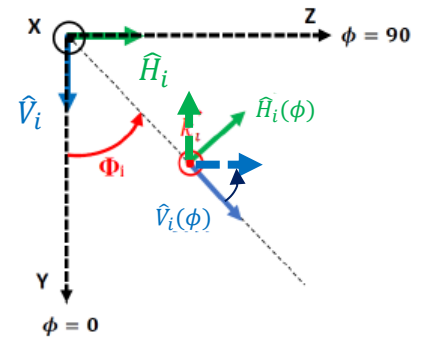
$$\hat{h}_s = -\sin \phi_s \hat{y} + \cos \phi_s \hat{z}$$

The 3rd angle β (being for now the same for incident and scattered wave) is used to define the wave polarization rotation around $\hat{k}_i$.

$$\begin{aligned} \hat{v}_{s,\beta} &= \cos \beta \hat{v}_s + \sin \beta \hat{h}_s \\ \hat{h}_{s,\beta} &= -\sin \beta \hat{v}_s + \cos \beta \hat{h}_s \end{aligned}$$

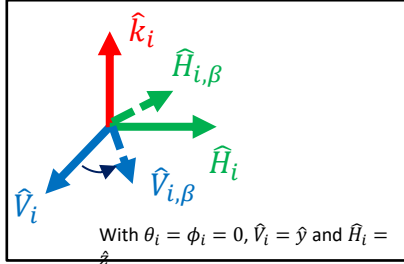


With $\theta_i = 0$, $\vec{v}$ and $\vec{h}$ stay in the YZ plan



MIDAS Incident direction and polarization rotation VS DDSCAT target orientation definitions

MIDAS



Which results in the below equations that we can also recover from DDSCAT User Guide equations (32)-(34) by using the established MIDAS-DDSCAT equivalence relations

Mathematically:

$$\hat{a}_1 = \hat{x}_{LF} \cos \Theta + \hat{y}_{LF} \sin \Theta \cos \Phi + \hat{z}_{LF} \sin \Theta \sin \Phi \quad (29)$$

$$\hat{a}_2 = -\hat{x}_{LF} \sin \Theta \cos \beta + \hat{y}_{LF} [\cos \Theta \cos \beta \cos \Phi - \sin \beta \sin \Phi] + \hat{z}_{LF} [\cos \Theta \cos \beta \sin \Phi + \sin \beta \cos \Phi] \quad (30)$$

$$\hat{a}_3 = \hat{x}_{LF} \sin \Theta \sin \beta - \hat{y}_{LF} [\cos \Theta \sin \beta \cos \Phi + \cos \beta \sin \Phi] - \hat{z}_{LF} [\cos \Theta \sin \beta \sin \Phi - \cos \beta \cos \Phi] \quad (31)$$

or, equivalently:

$$\hat{k}_s \hat{x}_{LF} = \hat{a}_1 \cos \Theta - \hat{a}_2 \sin \Theta \cos \beta + \hat{a}_3 \sin \Theta \sin \beta \quad (32)$$

$$\hat{v}_{s,\beta} \hat{y}_{LF} = \hat{a}_1 \sin \Theta \cos \Phi + \hat{a}_2 [\cos \Theta \cos \beta \cos \Phi - \sin \beta \sin \Phi] - \hat{a}_3 [\cos \Theta \sin \beta \cos \Phi + \cos \beta \sin \Phi] \quad (33)$$

$$\hat{H}_{s,\beta} \hat{z}_{LF} = \hat{a}_1 \sin \Theta \sin \Phi + \hat{a}_2 [\cos \Theta \cos \beta \sin \Phi + \sin \beta \cos \Phi] - \hat{a}_3 [\cos \Theta \sin \beta \sin \Phi - \cos \beta \cos \Phi] \quad (34)$$

$$\Rightarrow \hat{k}_s = \cos \theta_s \hat{x} + \sin \theta_s \cos \phi_s \hat{y} + \sin \theta_s \sin \phi_s \hat{z}$$

$$\Rightarrow \hat{v}_{s,\beta} = -\cos \beta \sin \theta_s \hat{x} + (\cos \beta \cos \theta_s \cos \Phi_s - \sin \beta \sin \Phi_s) \hat{y} + (\cos \beta \cos \theta_s \sin \Phi_s + \sin \beta \cos \Phi_s) \hat{z}$$

$$\Rightarrow \hat{H}_{s,\beta} = \sin \beta \sin \theta_s \hat{x} - (\sin \beta \cos \theta_s \cos \Phi_s + \cos \beta \sin \Phi_s) \hat{y} + (\cos \beta \cos \Phi_s - \sin \beta \cos \theta_s \sin \Phi_s) \hat{z}$$

MIDAS-DDSCAT angles equivalence:

$$\theta_{MIDAS} = \theta_{DDSCAT}$$

$$\text{so } \cos \theta_{DDSCAT} = \cos \theta_{MIDAS} \text{ and } \sin \theta_{DDSCAT} = \sin \theta_{MIDAS}$$

$$\varphi_{MIDAS} = \pi - \beta_{DDSCAT} \text{ so } \cos \beta_{DDSCAT} = -\cos \varphi_{MIDAS} \text{ and } \sin \beta_{DDSCAT} = \sin \varphi_{MIDAS}$$

$$\beta_{MIDAS} = \pi - \varphi_{DDSCAT} \text{ so } \cos \varphi_{DDSCAT} = -\cos \beta_{MIDAS} \text{ and } \sin \varphi_{DDSCAT} = \sin \beta_{MIDAS}$$

MIDAS Incident direction and polarization rotation VS DDSCAT target orientation definitions

Note that the equations detailed in the previous slides are used to calculate the components of the scattering matrix (S_{VV} , S_{VH} , S_{HV} and S_{HH}) by projecting the scattered fields calculated for a polarized incident direction onto the scattered polarizations vectors $\hat{V}_{s,\beta}$ and $\hat{H}_{s,\beta}$.

If we call the scattered fields calculated for incident wave polarizations $\hat{V}_i$ and $\hat{H}_i$, respectively E_v^s and E_h^s , the scattered field resulting from a polarized incident wave ($\beta = \beta_i = \beta_s \neq 0$) is the linear combination of E_v^s and E_h^s

$$\begin{aligned} E_{v,\beta}^s &= \cos \beta E_v^s + \sin \beta E_h^s \\ E_{h,\beta}^s &= -\sin \beta E_v^s + \cos \beta E_h^s \end{aligned}$$

If we combine the above equations, with the expressions of $\hat{V}_{s,\beta}$ and $\hat{H}_{s,\beta}$, we get:

$$S_{VV} = -\cos \beta \sin \theta_s E_{v,\beta}^{s,x} + (\cos \beta \cos \theta_s \cos \Phi_s - \sin \beta \sin \Phi_s) E_{v,\beta}^{s,y} + (\cos \beta \cos \theta_s \sin \Phi_s + \sin \beta \cos \Phi_s) E_{v,\beta}^{s,z}$$

$$S_{VH} = \sin \beta \sin \theta_s E_{v,\beta}^{s,x} - (\sin \beta \cos \theta_s \cos \Phi_s + \cos \beta \sin \Phi_s) E_{v,\beta}^{s,y} + (\cos \beta \cos \Phi_s - \sin \beta \cos \theta_s \sin \Phi_s) E_{v,\beta}^{s,z}$$

$$S_{HV} = -\cos \beta \sin \theta_s E_{h,\beta}^{s,x} + (\cos \beta \cos \theta_s \cos \Phi_s - \sin \beta \sin \Phi_s) E_{h,\beta}^{s,y} + (\cos \beta \cos \theta_s \sin \Phi_s + \sin \beta \cos \Phi_s) E_{h,\beta}^{s,z}$$

$$S_{HH} = \sin \beta \sin \theta_s E_{h,\beta}^{s,x} - (\sin \beta \cos \theta_s \cos \Phi_s + \cos \beta \sin \Phi_s) E_{h,\beta}^{s,y} + (\cos \beta \cos \Phi_s - \sin \beta \cos \theta_s \sin \Phi_s) E_{h,\beta}^{s,z}$$

MIDAS Incident direction and polarization rotation VS DDSCAT target orientation definitions

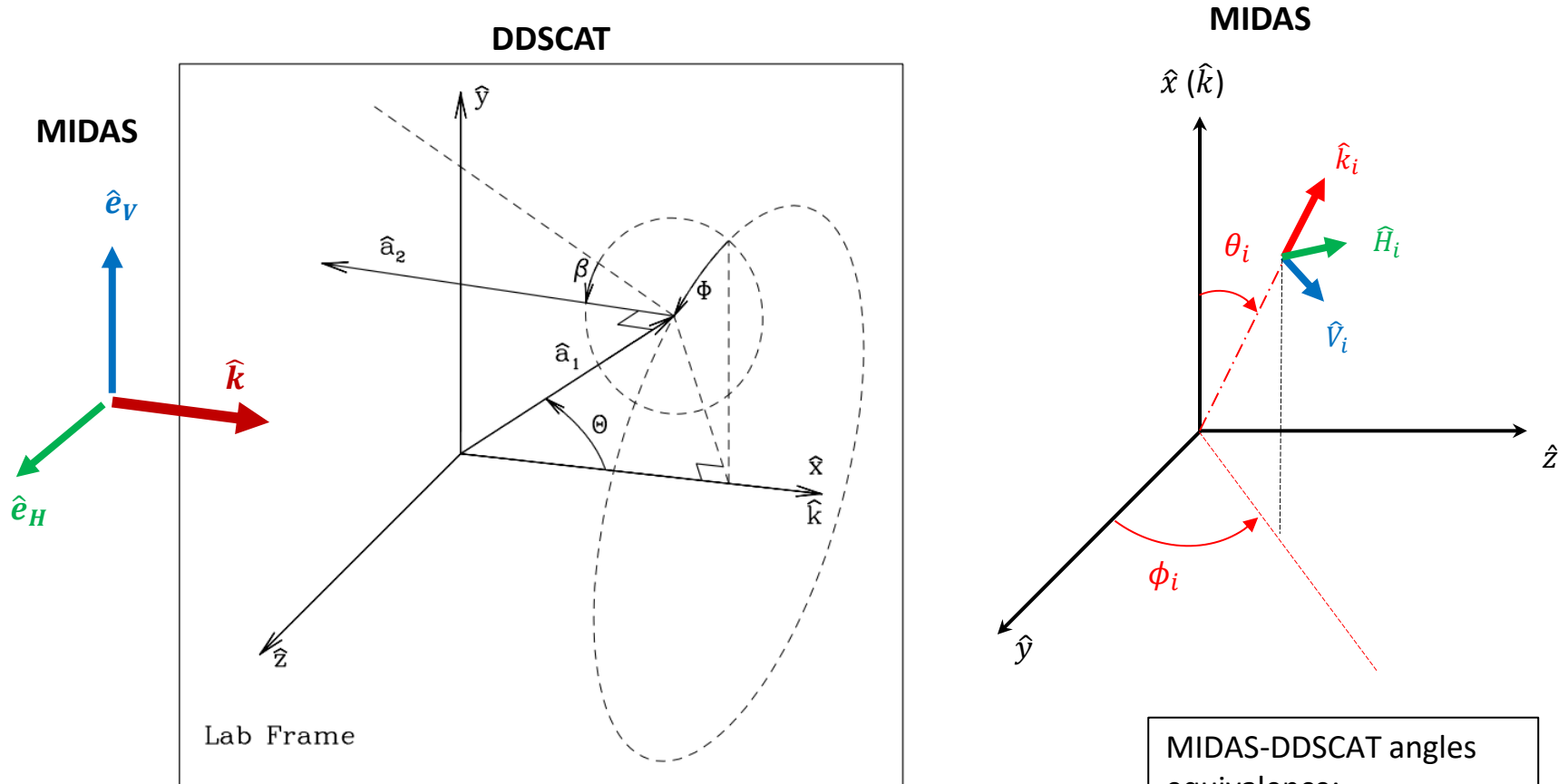


Figure 7: Target orientation in the Lab Frame. $\hat{x} = \hat{x}_{LF}$ is the direction of propagation of the incident radiation, and $\hat{y} = \hat{y}_{LF}$ is the direction of the real component (at $x_{LF} = 0, t = 0$) of the first incident polarization mode. In this coordinate system, the orientation of target axis $\hat{a}_1$ is specified by angles Θ and Φ . With target axis $\hat{a}_1$ fixed, the orientation of target axis $\hat{a}_2$ is then determined by angle β specifying rotation of the target around $\hat{a}_1$. When $\beta = 0$, $\hat{a}_2$ lies in the $\hat{a}_1, \hat{x}_{LF}$ plane.

MIDAS-DDSCAT angles
equivalence:

$$\theta_{MIDAS} = \theta_{DDSCAT}$$

$$\varphi_{MIDAS} = \pi - \beta_{DDSCAT}$$

$$\beta_{MIDAS} = \pi - \varphi_{DDSCAT}$$

MIDAS: MoM Integral-equation Decomposition for Arbitrarily-shaped Scatterers



**Iterative solver-
based (ISB)
DDA codes
vs
MIDAS**



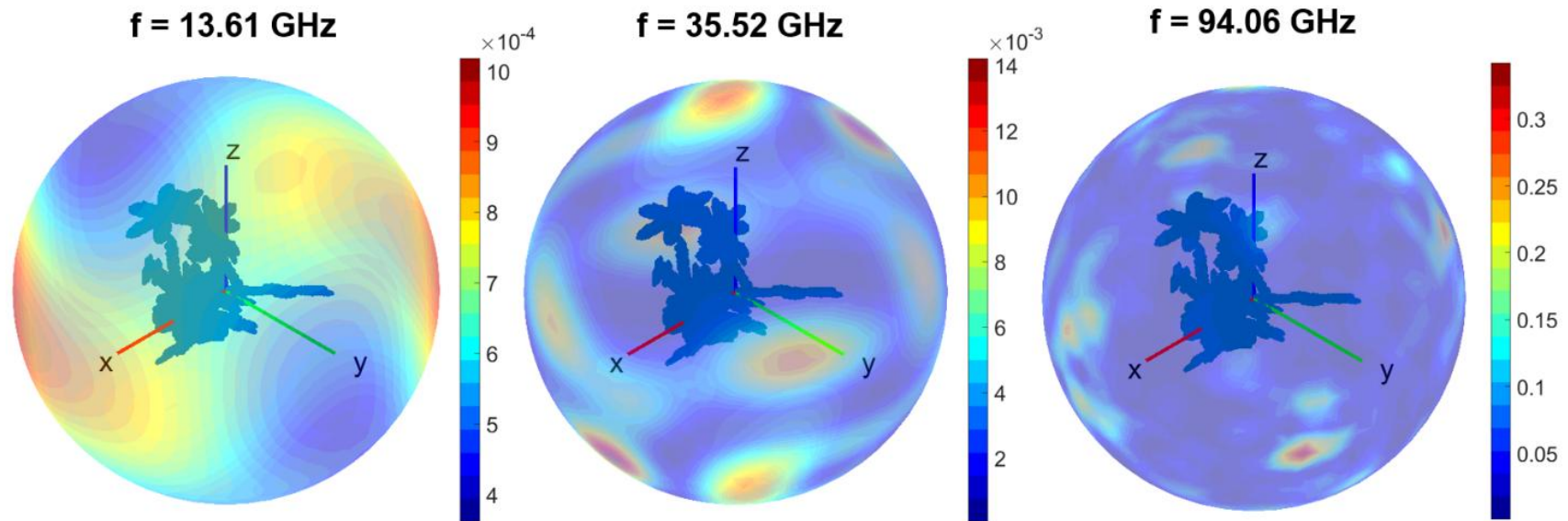
- Voxel meshing and piecewise constant electric field inside scatterer
- Validity Criteria: $|m|kd \leq 0.5$ for DDA or $\lambda/\sqrt{\mathcal{R}(\epsilon_r)} \leq 10$ for MoM
- Versatility regarding scatterer shape

→ Comparable accuracy levels and similar uncertainties for scatterers featuring high dielectric constants/contrasts

- Direct solver based (DSB) powerful DDM
- VIEM with possibility of different choices of mesh/basis functions

→ Higher computational efficiency particularly when considering orientation-averaging & higher amenability to accuracy enhancement

MIDAS & complex-shaped solid snow aggregates



Variation of the backscatter efficiency Q_{bks} of a0058 depending on the incident direction (θ_i, Φ_i) . One can see the impact of the frequency on Q_{bks} , as function of the particle anisotropy.

The variations of the backscatter efficiency, above, are obtained with the MPI-parallelized MIDAS for **2701 incidents directions (IDs)** (or, equivalently, target orientations) **at Ku, Ka and W bands** in **LESS THAN 20 minutes**, using **ONLY 2 computing nodes** (generally available without any queue time).

Such simulation is just unfeasible with the MPI-DDSCAT, with similar computational resources. MIDAS has a computational advantage over ADDA (known to be more efficient than DDSCAT)

MIDAS Ongoing & Perspectives



MIDAS

MIDAS has not only demonstrated significant advantage in computational efficiency over the DDA-based code DDSCAT, without compromising accuracy, but also great potential to address outstanding EM scattering challenges from solid and mixed-phase hydrometeors to comprehensively improve physical quantitative microwave precipitation retrievals.

Fenni et al., 2017. A computationally efficient 3-D full-wave model for coherent EM scattering from complex-geometry hydrometeors based on MoM/CBFM-enhanced algorithm. IEEE Transactions on Geoscience and Remote Sensing

Fenni et al, 2020. Evaluation of Higher-order Quadrature Schemes in Improving Computational Efficiency for Orientation-averaged Single-Scattering Properties of Nonspherical Ice Particles. JGR Atmospheres

MIDAS was developed with support from the National Aeronautics and Space Administration (80NSSC19K0687) within the Precipitation Measurement Missions Science Team (PMMST) program.



Open-Sourcing

VIEM/MoM enhancement

Optimize MPI parallelization

MPI Multilevel CBFM

Tests and Validation

Preferred orientation

ADDA

T-matrix

FEKO/CST

MIDAS: compile build and Run

MIDAS Private Unwatch 1

main 1 Branch 0 Tags

Go to file Add file Code

U-JPL\ifenni and U-JPL\ifenni adding analysis script 477634a · 4 minutes ago 165 Commits

docs	reorganizing example and tool folders	yesterday
examples	more reorg	23 minutes ago
src	minor for not showing progress when large numbers of bloc...	46 minutes ago
tools	adding analysis script	4 minutes ago
.gitignore	adding analysis script	4 minutes ago
LICENSE	Change license from GPL to Apache 2.0	last week
README.md	updating readme file	last week

README Apache-2.0 license

MIDAS

MIDAS (for MoM Integral-equation Domain-decomposition for Arbitrarily-shaped Scatterers) is an implementation of a numerically efficient 3D full-wave model for electromagnetic (EM) scattering from complex-shaped scatterers. The code is written in FORTRAN and is MPI (Message Passing Interface) parallelized. Several tools, written in MATLAB, are also provided to prepare and write the FORTRAN code inputs (shape, dielectric composition, simulation data files, etc.), or read and display its outputs (scattered fields, scattering matrices and efficiency factors, etc.).

/MIDAS_release/Run_Sims_p0004/

Name

..
inputs
midas
job_MIDAS_discover

'MIDAS_release/Run_Sims_p0004/inputs/

Name

diel
shape.dat
Simulation_data.dat

- git pull <https://github.jpl.nasa.gov/ifenni/MIDAS.git>
- Select (and update if needed) adequate makefile from the “makefiles” folder and place it in the src folder
- Load required modules. For example, for discover:
 module load comp/intel/2021.7.0
 module load mpi/impi/2021.7.0
- Build the executable midas
 make -f Makefile_discover modules
 make -f Makefile_discover
- Copy the executable to the “Run_Sims_*” folder and submit job. The submitted job will use the input files in the “Run_Sims_*/inputs” subfolder
- Note that the “Run_Sim” folder (can be renamed) must have the structure shown in the figure above (mainly input folder)

MIDAS: "Simulation_data" file at a glance

1.. Simulation_data.dat

0 10 20 30 40 50 60 70 80 90 100 110 120

SIMULATION DATAS :-

** General parameters of the simulation *****

NFreq = ! NFreq or NWave1
1
list ! minmax or list
Freqs (GHz) = ! Freqs (MHz/GHz/THz) or Wave1 (m/mm/um)
013.600

Number and values of frequencies or wavelengths. For a microwave simulation no need to change the order of magnitude, namely keep always GHz, and only change the frequency values.

** Parameters of the particles *****

shape_list & InOut Folders =

0

.

Type_p =

2p-0004

ap/L (m/mm/um)=

0.700283325

m_diffs =

fromonlymfile 1

'inputs/diel/m1.78_0.00133'

- Shape_list is set to 1 only if the code is reading the pathes to shapefiles from a file. So keep to 1 if no need for multiple shapes simulation
- The InOut Folder is set to be the output folder of the simulation (both single and multiple). Set to '.' if you would like to see the sim output folder in the current folder, otherwise set to something like "/gpfs/dnb32/ifenni" to avoid any disk quota issues
- Type and size (equivalent-volume sphere radius) of the particle and dielectric properties

** Parameters of the discretization *****

Sc (um/m) =

50

The cell size. **This input value will be considered only for spheres.** For shapes read from a file, Sc is calculated from ap and the number of cells read from the shape file

** Parameters of the applied methods *****

Nber_Methods =

1

methods_names =

CBFM-E

Number and names of used methods. Keep to 1 and CBFM-E for best performance of MIDAS for solid hydrometeors
For debug and validation purposes, this can be replaced with 2 : CBFM-E and MoM is 2 successive lines.

MIDAS: “Simulation_data” file at a glance

**** Parameters of transmitters/receivers *******

```
quad_type =
sd/sd
Ninc (sd/lb) =
482
Nscat (sd/lb) =
482
tr_config (un/gl) =
theta =
0 180 19
phi =
0 360 37
Rec_config (un/gl) =
theta =
0 180 181
phi =
0 360 25
Nbeta (0-2pi polar. rotation)=
1
wr_Si =
1
wr_Qi =
1
```

- This section focuses on the Incident/Scattering (Tx/Rx) directions and polarization rotation information.
- For both Tx and Rx configuration, the distribution type can be set to un (for uniform step in theta and phi, the adaptive quadrature is used for the orientation averaging for this type of distribution), sd (spherical design), lb (Lebedev quadrature) and rf (read from file).
- If the quadrature type is set to sd, lb or rf, the sections tr_config and Rec_config are ignored, and the only information that will be used if the values of Ninc and Nsca.
- For sd, Ninc and Nsca can be set to [62, 72, 86, 98, 114, 128, 146, 163, 182, 201, 222, 243, 266, 289, 314, 339, 366, 393, 422, 451, 482, 513, 546, 579, 614, 649, 686, 723, 762, 801, 842, 883, 926, 3282]
- For lb, Ninc and Nsca can be set to [86, 110, 146, 170, 194, 230, 266, 302, 350, 434, 590, 770, 974, 1202, 1454, 1730, 2030, 2354, 2702, 3074, 3470, 3890, 4334, 4802, 5294, 5810]
- Nbeta is the number of requested polarizations. For example, for a polarization rotation with a step of 30 degrees, Nbeta = 13. Put Nbeta to 1 if no polarization rotation is considered, and scattering matrices are only calculated for V and H polarizations.
- wr_Si and wr_Qi take 0 or 1 and define if the S matrices and efficiencies factors will be written for each incident direction. If wr_Si and wr_Qi are set to 0, the S and Q folders will be created and populated only if the number of incident directions is equal to 1.

**** Parameters of the CBFM *******

```
div_type = (CSH or SPH)
CSH
NBlock (avgNcells) =
2000
IPWs (type & number)
sd
422
```

- The division into blocks for the CBFM: the type should be set to ‘CSH’ (complex-shaped) for arbitrary scatterers and to ‘SPH’ for pseudo-spherical objects
- Nblock refers to the average number of cells per block. 2000 is a value that works well with most cases. Increasing this parameter improves the CBFM compression but has a memory usage cost.
- IPWs (type & number) : the type and number of incident plane waves used to calculate the electric field for each CBFM block. Our analysis shows that a spherical design (sd) distribution with 422 nodes or less works well with most of the solid-ice particles cases. The user can select a lower number for better computational efficiency.

MIDAS: "Simulation_data" file

1.. Simulation_data.dat

```
0          10          20          30          40          50          60          70          80          90          100          110          120
2000

** Scattering Solution Type/Outputs *****
CextIntFields =
0
FarFieldApp =
1
R (m) (used if FFA=0) =
500

** Optional Saved Solution Elts *****
save_Zc&Ctot =
0
save_Eint =
0
10E4
save_Einc =
0
```

- The last 2 sections include various scattering solution parameters that can be set to the default values as shown here.
- CextIntFields may be set to 1 if the user would like to calculate extinction and absorption cross sections from internal fields. The default is calculating both quantities from scattered fields
- Setting FarFieldApp to 0 will cancel the far field approximation and the code will calculate and output scattered fields at R instead of scattering matrices.
- The 3 last parameters are to set to 1 if the user wants to output and save the compressed matrix, the CBFs, the total fields inside the particle (with a condition Nmax), and the incident fields.

MIDAS: Structure of a Sim folder

Run_Sims_p0004 > expected_outputs

Name

p-0004-ap=0.700283325mm
output.33200534
Sim_MIDAS_05-21-2024_11h45

Run_Sims_p0004 > expected_outputs > p-0004-ap=0.700283325mm >

Name

Q_files
S_files
Blocks
Cells
IncScattDirs
qtable_CBFM-E
Simulation_data

- **Simulation_data.dat**: input parameter copied to simulation folder from input folder
- **IncScattDirs.dat** describes the Incident/scattering directions used for the simulation. The same file can be used as input file with the **quad_type** set to 'rf'
- Cells.dat and Blocks.dat are output geometry files that describe the discretization (or meshing) of the particle and the division into blocks that was used to apply the Characteristic Basis Function Method (CBFM)
- Sim_MIDAS_*. and output.* are simulation output files used to understand how the simulation progressed, look at the computational cost, calculation time, conditioning of the matrix
- qtable_CBFM-E.dat is an output file that has all **averaged** scattering efficiency factors (Q) for each simulated frequency
- The folders Q_files and S_files include the detailed calculated scattering efficiency factors Q and scattering matrices (S for 2 co-polarizations S_{vv}, S_{hh} and for 2 cross-polarizations S_{vh} and S_{hv}) for each incident direction and each frequency.

MIDAS: Scattering efficiency factors Q

Examples\Run_Sims_p0004\

p-0004-ap=0.700283325mm>qtable_CBFM-E.dat

1.. qtable_CBFM-E.dat

```

0      10      20      30      40      50      60      70      80      90      100
Number of Cells = 11508
Results averaged over 703 incident directions (aq) :
Theta_i_min = 0.00; Theta_i_max = 180.00; NTheta_i = 19
Phi_i_min = 0.00; Phi_i_max = 360.00; NPhi_i = 37
Results calculated with 4525 scattering directions (aq) :
Theta_s_min = 0.00; Theta_s_max = 180.00; NTheta_s = 181
Phi_s_min = 0.00; Phi_s_max = 360.00; NPhi_s = 25
      freq      aeff      wave      Q_ext      Q_abs      Q_scatt      Q_bk      g(1)=<cos>      Ncells
1.3600E+10  7.0028E+02  2.2059E+04  1.7273E-03  6.7804E-04  1.0509E-03  1.1027E-04  4.2052E-02  11508
    
```

Examples\Run_Sims_p0004\expected_outputs\p-0004-ap=0.700283325mm>>Q_files>Qidtable_13.60GHz_CBFM-E.dat

1.. qtable_CBFM-E.dat

2.. Qidtable_13.60GHz_CBFM-E.dat

```

0      10      20      30      40      50      60      70
      theta      phi      Q_ext      Q_abs      Q_scatt      Q_bk      g(1)=<cos>
      0.00      0.00  2.4151E-03  9.1879E-04  1.4990E-03  1.8775E-04  1.8495E-04
      10.00      0.00  2.3835E-03  8.3939E-04  1.4781E-03  1.8301E-04  1.8857E-03
      20.00      0.00  2.2934E-03  7.4834E-04  1.4184E-03  1.7003E-04  6.8390E-03
      30.00      0.00  2.1570E-03  6.5676E-04  1.3285E-03  1.5197E-04  1.4607E-02
      40.00      0.00  1.9923E-03  5.7580E-04  1.2207E-03  1.3259E-04  2.4488E-02
      50.00      0.00  1.8204E-03  5.1527E-04  1.1091E-03  1.1498E-04  3.5485E-02
      60.00      0.00  1.6618E-03  4.8244E-04  1.0069E-03  1.0096E-04  4.6426E-02
      70.00      0.00  1.5346E-03  4.8116E-04  9.2561E-04  9.1167E-05  5.5867E-02
      80.00      0.00  1.4526E-03  5.1147E-04  8.7348E-04  8.5490E-05  6.2367E-02
      90.00      0.00  1.4243E-03  5.6962E-04  8.5555E-04  8.3643E-05  6.4686E-02
     100.00      0.00  1.4526E-03  6.4861E-04  8.7349E-04  8.5490E-05  6.2366E-02
     110.00      0.00  1.5346E-03  7.3896E-04  9.2561E-04  9.1167E-05  5.5861E-02
     120.00      0.00  1.6618E-03  8.2991E-04  1.0069E-03  1.0096E-04  4.6426E-02
     130.00      0.00  1.8204E-03  9.1058E-04  1.1091E-03  1.1498E-04  3.5490E-02
     140.00      0.00  1.9923E-03  9.7131E-04  1.2207E-03  1.3259E-04  2.4478E-02
    
```

- theta and phi here describe the incident direction (θ_i, Φ_i)
- Each line corresponds to one of the 703 incident directions over which the averaged efficiency factors were calculated.
- We have a Qidtable file for each simulated frequency

MIDAS: Scattering matrices S

Examples\Run_Sims_p0004\

p-0004-ap=0.700283325mm\S_files\ Smtable_13.60GHz_kt0001_CBFM-E.dat

1.. Smtable_13.60GHz_kt0001_CBFM-E.dat

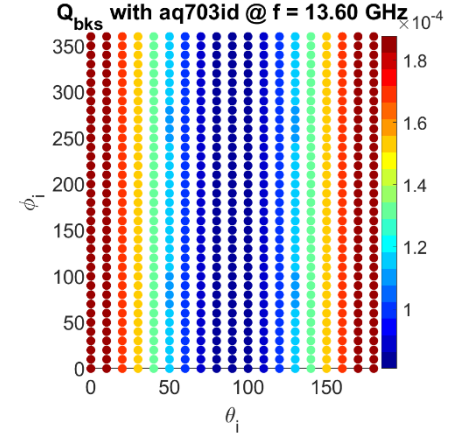
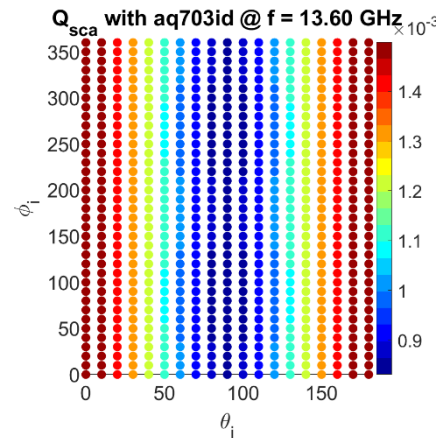
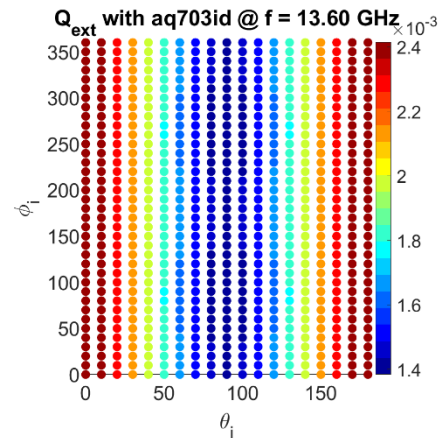
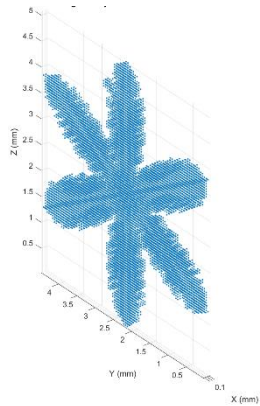
0	10	20	30	40	50	60	70	80	90	100	110
THETA = 0.0000; PHI = 0.0000; BETA = 0.0000											
theta	phi	Re(Svv)	Im(Svv)	Re(Svh)	Im(Svh)	Re(Shv)	Im(Shv)	Re(Shh)	Im(Shh)		
0.0000	0.0000	0.4805E-02	0.2361E-04	0.5636E-05	0.5504E-07	0.5636E-05	0.5504E-07	0.4888E-02	0.2444E-04		
1.0000	0.0000	0.4804E-02	-0.2986E-04	0.5636E-05	0.4661E-07	0.5635E-05	0.3948E-07	0.4888E-02	-0.2997E-04		
2.0000	0.0000	0.4801E-02	-0.8322E-04	0.5637E-05	0.3822E-07	0.5633E-05	0.2397E-07	0.4887E-02	-0.8431E-04		
3.0000	0.0000	0.4796E-02	-0.1364E-03	0.5639E-05	0.2985E-07	0.5630E-05	0.8504E-08	0.4885E-02	-0.1386E-03		
4.0000	0.0000	0.4788E-02	-0.1893E-03	0.5640E-05	0.2150E-07	0.5625E-05	-0.6904E-08	0.4883E-02	-0.1927E-03		
5.0000	0.0000	0.4779E-02	-0.2419E-03	0.5643E-05	0.1315E-07	0.5619E-05	-0.2225E-07	0.4880E-02	-0.2467E-03		
6.0000	0.0000	0.4767E-02	-0.2941E-03	0.5645E-05	0.4799E-08	0.5611E-05	-0.3751E-07	0.4876E-02	-0.3005E-03		
7.0000	0.0000	0.4753E-02	-0.3458E-03	0.5648E-05	-0.3575E-08	0.5602E-05	-0.5270E-07	0.4872E-02	-0.3541E-03		
8.0000	0.0000	0.4737E-02	-0.3969E-03	0.5652E-05	-0.1198E-07	0.5591E-05	-0.6779E-07	0.4867E-02	-0.4074E-03		
9.0000	0.0000	0.4719E-02	-0.4474E-03	0.5656E-05	-0.2042E-07	0.5579E-05	-0.8278E-07	0.4861E-02	-0.4606E-03		
10.0000	0.0000	0.4699E-02	-0.4973E-03	0.5660E-05	-0.2892E-07	0.5565E-05	-0.9766E-07	0.4855E-02	-0.5134E-03		
11.0000	0.0000	0.4677E-02	-0.5463E-03	0.5665E-05	-0.3747E-07	0.5550E-05	-0.1124E-06	0.4848E-02	-0.5660E-03		
12.0000	0.0000	0.4653E-02	-0.5946E-03	0.5670E-05	-0.4609E-07	0.5533E-05	-0.1270E-06	0.4840E-02	-0.6182E-03		
13.0000	0.0000	0.4627E-02	-0.6419E-03	0.5676E-05	-0.5479E-07	0.5515E-05	-0.1415E-06	0.4832E-02	-0.6701E-03		
14.0000	0.0000	0.4599E-02	-0.6883E-03	0.5682E-05	-0.6358E-07	0.5495E-05	-0.1559E-06	0.4823E-02	-0.7215E-03		
15.0000	0.0000	0.4569E-02	-0.7337E-03	0.5688E-05	-0.7245E-07	0.5474E-05	-0.1700E-06	0.4813E-02	-0.7726E-03		
16.0000	0.0000	0.4537E-02	-0.7779E-03	0.5695E-05	-0.8143E-07	0.5451E-05	-0.1840E-06	0.4803E-02	-0.8233E-03		
17.0000	0.0000	0.4503E-02	-0.8211E-03	0.5702E-05	-0.9051E-07	0.5427E-05	-0.1978E-06	0.4792E-02	-0.8735E-03		
18.0000	0.0000	0.4468E-02	-0.8630E-03	0.5709E-05	-0.9970E-07	0.5401E-05	-0.2114E-06	0.4781E-02	-0.9233E-03		
19.0000	0.0000	0.4431E-02	-0.9038E-03	0.5717E-05	-0.1090E-06	0.5374E-05	-0.2248E-06	0.4769E-02	-0.9726E-03		
20.0000	0.0000	0.4392E-02	-0.9432E-03	0.5725E-05	-0.1184E-06	0.5345E-05	-0.2380E-06	0.4757E-02	-0.1021E-02		
21.0000	0.0000	0.4351E-02	-0.9812E-03	0.5733E-05	-0.1280E-06	0.5315E-05	-0.2509E-06	0.4744E-02	-0.1070E-02		
22.0000	0.0000	0.4309E-02	-0.1018E-02	0.5742E-05	-0.1377E-06	0.5283E-05	-0.2636E-06	0.4731E-02	-0.1117E-02		
23.0000	0.0000	0.4265E-02	-0.1053E-02	0.5751E-05	-0.1475E-06	0.5249E-05	-0.2760E-06	0.4717E-02	-0.1164E-02		
24.0000	0.0000	0.4220E-02	-0.1087E-02	0.5760E-05	-0.1574E-06	0.5214E-05	-0.2882E-06	0.4702E-02	-0.1211E-02		
25.0000	0.0000	0.4173E-02	-0.1119E-02	0.5769E-05	-0.1675E-06	0.5177E-05	-0.3000E-06	0.4688E-02	-0.1257E-02		
26.0000	0.0000	0.4125E-02	-0.1150E-02	0.5779E-05	-0.1777E-06	0.5139E-05	-0.3115E-06	0.4673E-02	-0.1302E-02		
27.0000	0.0000	0.4075E-02	-0.1179E-02	0.5788E-05	-0.1880E-06	0.5099E-05	-0.3227E-06	0.4657E-02	-0.1347E-02		
28.0000	0.0000	0.4024E-02	-0.1206E-02	0.5798E-05	-0.1984E-06	0.5058E-05	-0.3336E-06	0.4641E-02	-0.1391E-02		

- theta and phi here describe the scattering direction (θ_i, Φ_i). So, each line corresponds to the real and imaginary parts of Sv_v, Sv_h, Sh_v and Sh_h for each scattering direction, for the incident direction kt0001
- We have one Smtable file for each of the 703 incident directions and each simulated frequency.

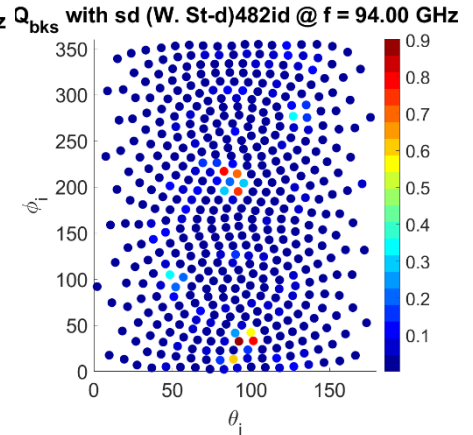
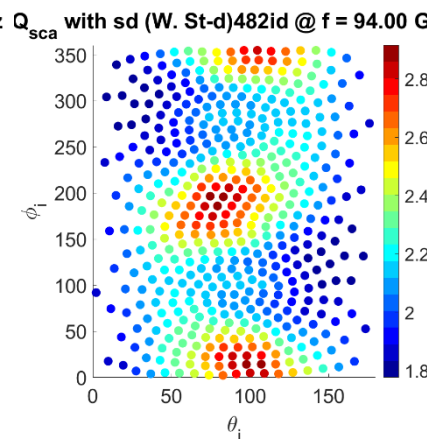
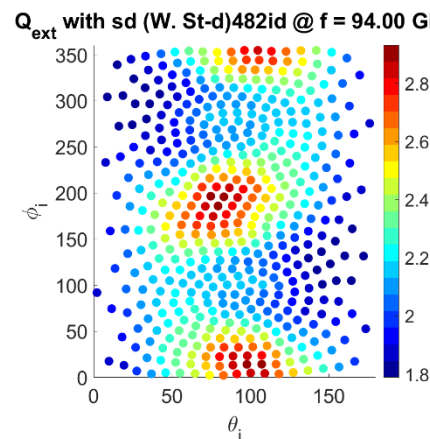
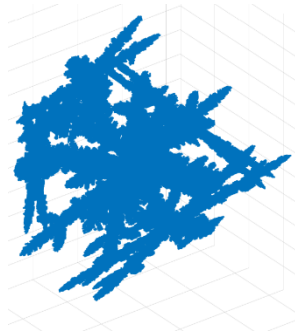
MIDAS: Examples

- p-0004: Type_p = 2p-0004 & ap = 0.700283325 mm : geometry read from shapefile representing a pristine from the OpenSSP database. Type_p starts with 2 whenever the shape is arbitrary and read from a shapefile. The text after 2 is used to name the output simulation folder

Similar simulation_data file can be used to run any homogeneous pristine particle by updating the frequency, the dielectric properties, the equivalent-volume sphere radius, the Tx/Rx configuration ...



- a-0006: Type_p = 2a-0006 & ap = 1.614011963 mm : geometry read from shapefile representing an aggregate from the OpenSSP database. Similar input properties as p-0004. The main difference with the previous simulation (other than geometry/frequency properties) is the use of **spherical design for the Tx/Rx definition**



MIDAS: Examples

- **sphere**: Type_p = 1 & ap = 0.8 mm : with type_p=1, the code build a sphere internally and doesn't look for a shapefile. For the example included in the starter kit, we calculate scattering by a sphere of radius 0.8 mm for 2 frequencies 013.6 and 35.5 GHz. Below is the comparison with Mie for f=13.6 GHz

